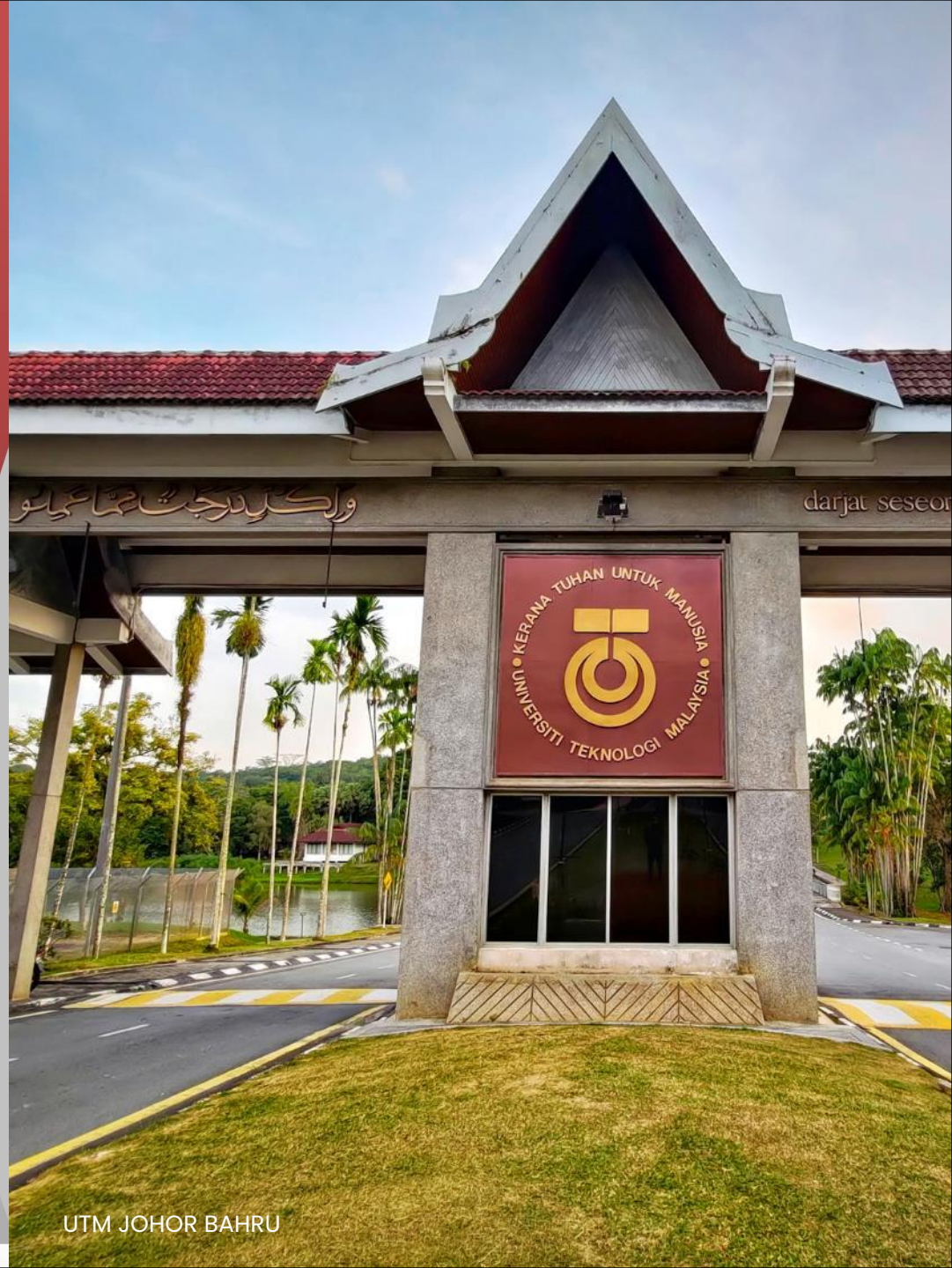
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SECJ 3483 – WEB TECHNOLOGY

TOPIC 01 – Introduction

innovative • entrepreneurial • global



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Objectives

By the end of this lesson, students should be able to:

- Explain the fundamental concepts of modern web technologies, including Frontend, Backend, Database and APIs in web application development.
- Differentiate between various frontend and backend technologies, understanding their advantages, use cases, and popular frameworks.
- Describe the importance of RESTful APIs and how HTTP methods (GET, POST, PUT, DELETE) facilitate data exchange between frontend and backend.

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Components of Modern Web Technologies

Frontend

Responsible for user interaction and experience.

Backend

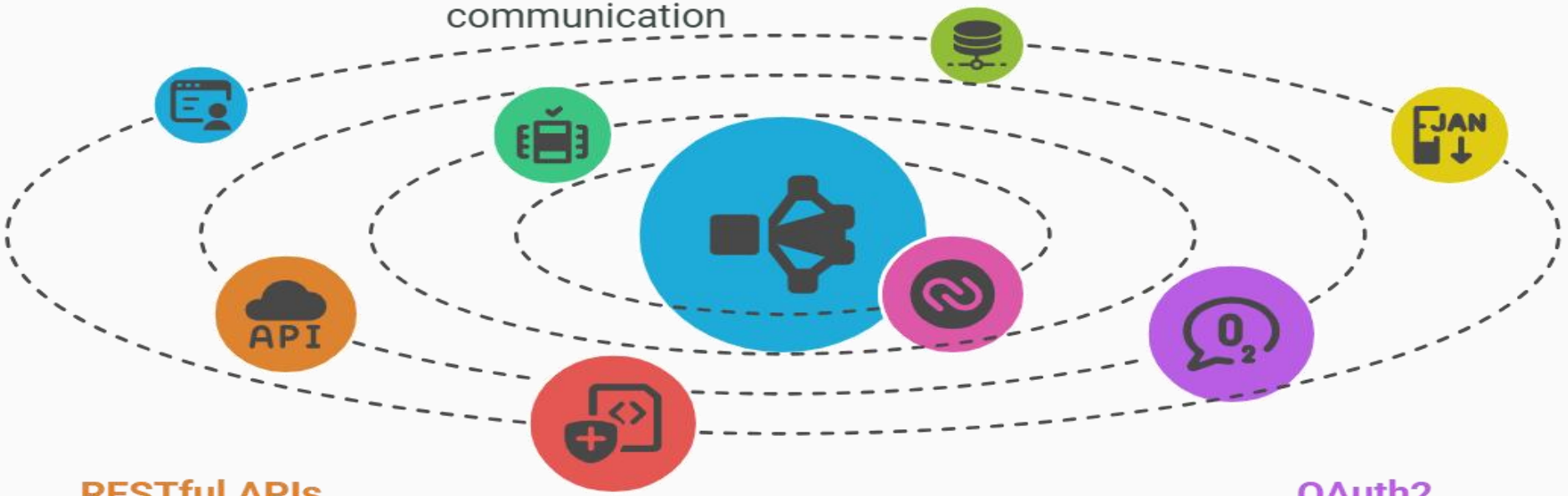
Responsible for data processing, authentication, business logic, and API communication

Database

Stores and organizes data

JSON

Lightweight data format for APIs



RESTful APIs

Bridges frontend and backend

Security

Protects applications from threats

JWT

Secures API authentication

OAuth2

Enables third-party authentication

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Frontend Technologies (Client-Side Development)

CSS

Styles web pages by controlling layout, colors, and animations to enhance visual appeal.

JavaScript

Adds interactivity to web pages through event handling and DOM manipulation.



HTML

Defines the structure of web pages, organizing content into elements like text, images, links, and forms.

CSS Frameworks & UI Libraries.

Provides tools for responsive design and prebuilt UI components, streamlining development.

Frontend Frameworks for Modern Web Development

React.js

Utilizes a virtual DOM for fast rendering, making it perfect for single-page applications (SPAs).

Vue.js

Known for its lightweight and easy-to-learn structure, Vue.js is component-based and ideal for quick development.



Angular

An enterprise-grade framework with robust two-way data binding, suitable for complex applications.

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Core Frontend Features

API Communication

Facilitates data exchange between client and server using Axios and Fetch API.

State Management

Involves managing application state efficiently with tools like Vuex, Pinia, and Redux.

Responsive Design

Ensures web applications are mobile-first and adaptable using CSS Flexbox, Grid, and Media Queries.



Component-based UI

Focuses on building reusable UI elements using frameworks like Vue, React, and Angular.

Progressive Web Applications

Enables offline functionality and enhanced performance with service workers.

Backend Frameworks (Server-Side Development)

Django

A secure, high-level framework with an admin panel.

Node.js

A JavaScript runtime for fast, scalable backends.

Spring Boot

An enterprise-grade framework for large-scale applications.



PHP Slim

A lightweight framework ideal for REST APIs.

ASP.NET

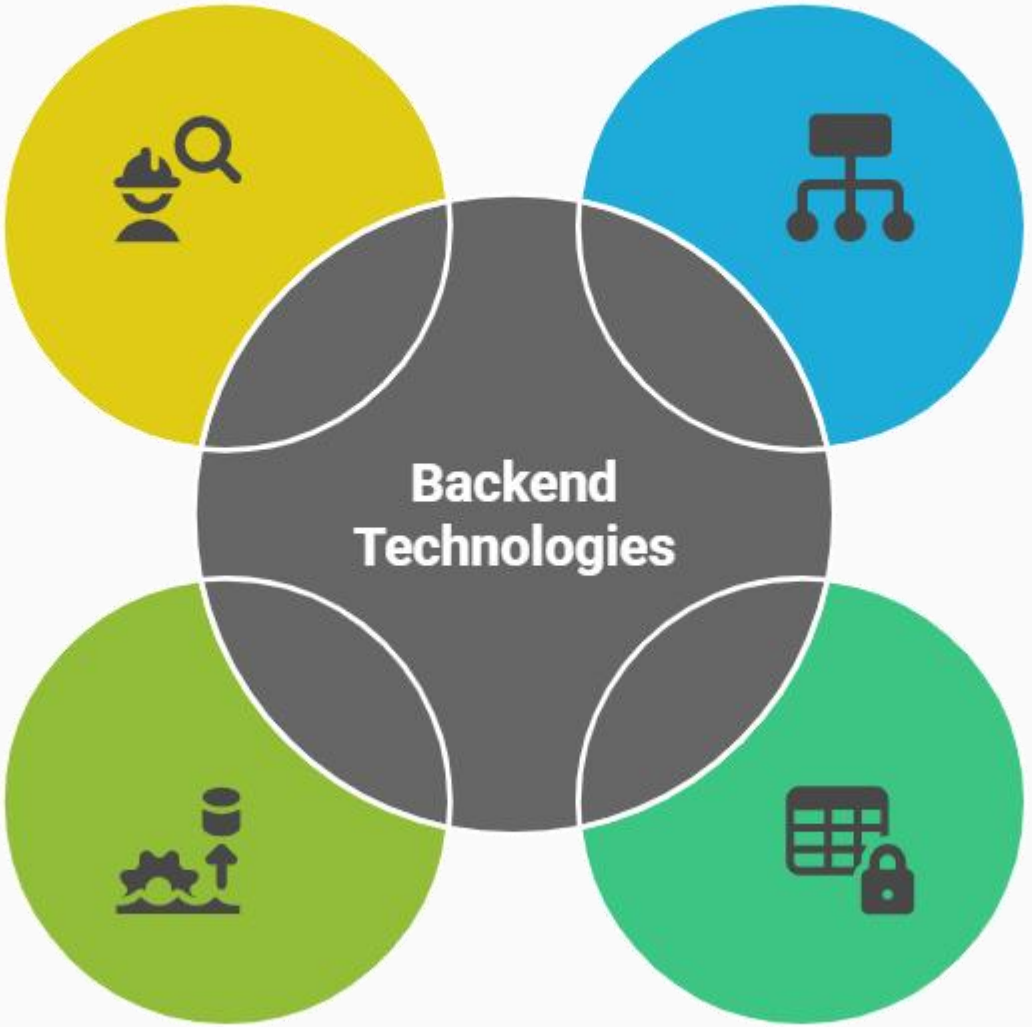
A backend framework integrated with the Microsoft stack.

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Core Backend Functionalities

Business Logic & Data Processing

Handles application workflows and computational tasks.



Routing & HTTP Handling

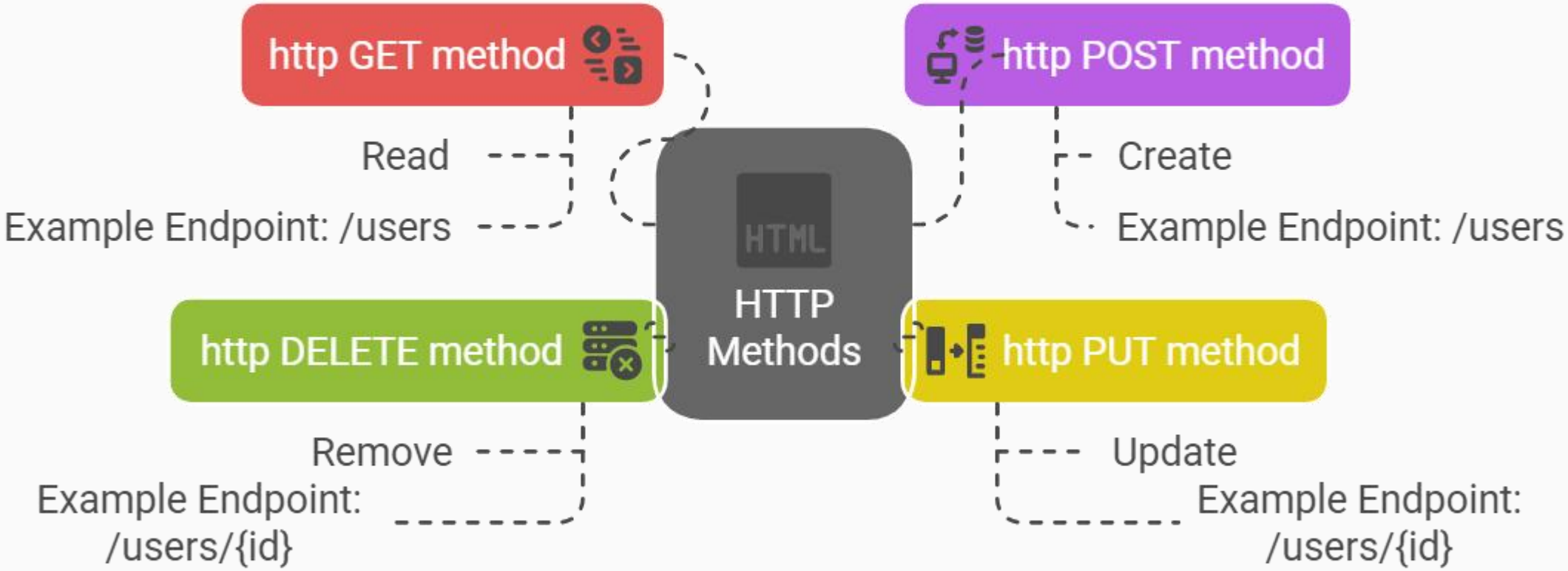
Manages URL endpoints for efficient data processing.

Authentication & Authorization

Secures access using session-based, JWT, and OAuth methods.

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RESTful APIs (Communication Between Frontend & Backend)



Uses different http methods to match CRUD operations

Storage & Mgmnt:
Which database technology to use for a web application?



Relational Database (MySql, SqLite)
Suitable for applications requiring structured data and complex queries.

Non-relational Database (MongoDB. Firebase)
Ideal for applications needing flexibility and scalability with unstructured data.

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Security in Web Applications

XSS Prevention

Escape user input to prevent the injection of malicious JavaScript into webpages.

SQL Injection Prevention

Use prepared statements and Object-Relational Mappers (ORMs) to thwart SQL query injections.



CSRF Prevention

Implement CSRF tokens to secure against unauthorized user actions.

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Authentication (JWT & OAuth2)

Use JWT

Suitable for applications requiring custom authentication solutions.



Use OAuth2

Ideal for applications needing third-party login integration.

Authentication ensures secure access to applications.

Cloud Technologies (Hosting & Deployment) for Scalable and Secure Deployment

Key Deployment Technologies

Essential tools and methods for efficient application deployment.

Provider Features

Major cloud providers offering unique services for various needs.



Scalability and Security

Ensuring applications can grow and remain secure in the cloud.